Table 3. Comparison of temporal approaches for the random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), Thailand, 2016–2023.

Temporal approach	Temporal structure selected in[b]	Number of parameters	Random parameters / heterogeneity in means / in variances	Variables with temporally varying parameters	LL(β)	AIC	BIC	LR test against the fully unconstrained model, χ^2 (df)[c]	Confidence of rejection
S1: single-motorcycle crashes (two-year periods); N = 7,325									
Fully unconstrained (separate period models)[a]	–	92	7 / 5 / 0	all	–4,510.10	9,204.2	9,838.9	–	–
Partially constrained (LR tests, 90%)	Fixed-parameter logit	52	6 / 3 / 0	23	–4,526.94	9,157.9	9,516.6	33.7 (40)	0.251
Adaptive fused (90% rule)	Fixed-parameter logit	45	7 / 2 / 1	16	–4,534.14	9,158.3	**9,468.7**	48.1 (47)	0.571
S2: motorcycle and another vehicle type; N = 21,371									
Fully unconstrained (separate period models)[a]	–	239	15 / 15 / 0	all	–12,010.21	24,498.4	26,403.2	–	–
Partially constrained (LR tests, 90%)	Fixed-parameter logit	85	8 / 3 / 0	36	–11,979.04	24,128.1	24,805.5	–62.3 (154)	0.000
Adaptive fused (90% rule)	Random parameters model	56	8 / 3 / 1	10	–12,052.03	24,216.1	**24,662.4**	83.6 (183)	0.000
S3: single vehicle, no motorcycle; N = 73,731									
Fully unconstrained (separate period models)[a]	–	238	16 / 14 / 4	all	–23,555.97	47,587.9	49,779.5	–	–
Partially constrained (LR tests, 90%)	Fixed-parameter logit	83	7 / 3 / 1	32	–23,590.45	47,346.9	48,111.2	69.0 (155)	0.000
Adaptive fused (90% rule)	Fixed-parameter logit	65	8 / 3 / 1	12	–23,646.87	47,423.7	**48,022.3**	181.8 (173)	0.692
S4: multi-vehicle, no motorcycle; N = 33,646									
Fully unconstrained (separate period models)[a]	–	215	15 / 12 / 4	all	–12,541.73	25,513.5	27,324.5	–	–
Partially constrained (LR tests, 90%)	Random parameters model	81	8 / 3 / 1	26	–12,598.28	25,358.6	26,040.9	113.1 (134)	0.095
Adaptive fused (90% rule)	Random parameters model	72	8 / 3 / 1	24	–12,617.43	25,378.9	**25,985.4**	151.4 (143)	0.701

Notes. All models: random parameters normally distributed, 2,000 Halton draws; random parameters and heterogeneity terms searched with 100 draws and confirmed with 2,000 draws; non-significant parameters removed (90% level). Lowest BIC within each subset among models not rejected at the 90% level in bold. Partially constrained: adjacent periods merged when a likelihood ratio test does not reject equal parameters at the 90% level. Adaptive fused: sparsest penalty of the adaptive fused lasso whose structure is not rejected at the 90% level against the fully unconstrained model (LR < $\chi^2_{0.90}$(df)). Variables with temporally varying parameters count the variables (including the constant) whose parameter differs between at least two periods.

[a] Separate random parameters models for each period (two-year periods for S1, yearly periods for S2–S4); parameter counts and LL(β) are summed over the period models, and AIC and BIC are computed from these sums with the total number of observations.

[b] Each approach selects its temporal structure either in the fixed-parameter binary logit (Stage 1, Table 2) or inside the random parameters model; the stage with the lower BIC among models not rejected at the 90% level is reported. Both stages for both approaches, and every candidate structure estimated, are given in Table D1.

[c] $\chi^2 = -2[LL(\beta)_{\text{temporal model}} - \Sigma\, LL(\beta)_{\text{period models}}]$, df = difference in the number of parameters. Because the period models are specified separately, the test is not strictly nested and χ^2 can be negative when the temporal model fits better than the period models with fewer parameters; such a model is not rejected. Confidence of rejection is the χ^2 cumulative probability; values below 0.90 mean that the model is not rejected at the 90% level.